



Growth & Development in Pediatrics

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- **Growth**: Increase in size of the body or quantitative growth
- **Development**: Maturation of function or qualitative growth

Growth

Stages of Growth

- Ovum (0 – 14 days)
- Embryo (2 – 9 weeks)
- Fetus (9 weeks – Birth)
- Newborn (0 – 28 days of life)
- Infant (1st year of life)
- Toddler (1 – 3 years)
- Pre-school child (3 – 5 years)
- School child (5 – 9 years)
- Adolescent (10 – 19 years)

Anthropometry

- Greek: '*anthropos*' – human
'*metros*' – measure
- It is the assessment of growth performed by measuring various growth parameters using appropriate equipment and comparing them to age-appropriate reference standards.

Parameters of Anthropometry

- Weight
- Length/Height
- Head circumference
- Chest circumference
- Mid-arm circumference
- Upper segment: Lower segment ratio
- Arm Span

Measurement of Weight



Measurement of Length



Measurement of Height



Measurement of Head Circumference



Birth Weight Related Terminologies

- 1.5kg – 2.5 kg: Low Birth Weight (LBW)
- 1kg – 1.5kg: Very Low Birth Weight (VLBW)
- <1kg: Extremely Low Birth Weight (ELBW)

Gestational Age Related Terminologies

- <28 weeks: Extreme Pre-term
- 28 – 32 weeks: Very Preterm
- 32 – 37 weeks: Pre-term
- 37 – 42 weeks: Term
- >42 weeks: Post-term

Weight

- Newborn loses upto 10% of its birth weight in the 1st 5-7 days and regains birthweight by 10 days
- 1st 3 months: 200 gms/week (30 gms/day) (after regaining birth weight)
- Next 3 months: 150 gms/week (20 gms/day)
- Next 6 months: 100 gms/week (10 gms/day)
- Birth weight doubles by 4 months, triples by 12 months and quadruples by 4 years
- Thereafter, 2kg per year till 6 years and thereafter 3 kg per year till puberty

Formula for Weight

S.no.	Age	Weight (Weech's Formula)
1.	3 – 12 months	$[\text{Age (months)} + 9] \div 2$
2.	1 – 6 years	$[\text{Age (years)} \times 2] + 8$
3.	7 – 12 years	$[\{\text{Age(years)} \times 7\} - 5] \div 2$

Height

- Birth – 50 cm
- 1 year – 75 cm
- 2 years – 87 cm
- 3 years – 94 cm
- 4 years – 100 cm
- Thereafter, 6cm per year till puberty
- Birth length triples by 12 years
- 2 – 12 years: $[\text{Age (years)} \times 6] + 77$

Head Circumference

- At birth – 33 to 35 cm
- Increases by:
 - 2 cm/month in the 1st 3 months
 - 1 cm/month in the next 3 months
 - 0.5 cm/month in the next 6 months

Chest Circumference

- At birth: Head circumference $>$ Chest circumference by 1 cm
- At 1 year: Head circumference = Chest circumference
- Thereafter: Chest circumference $>$ Head circumference

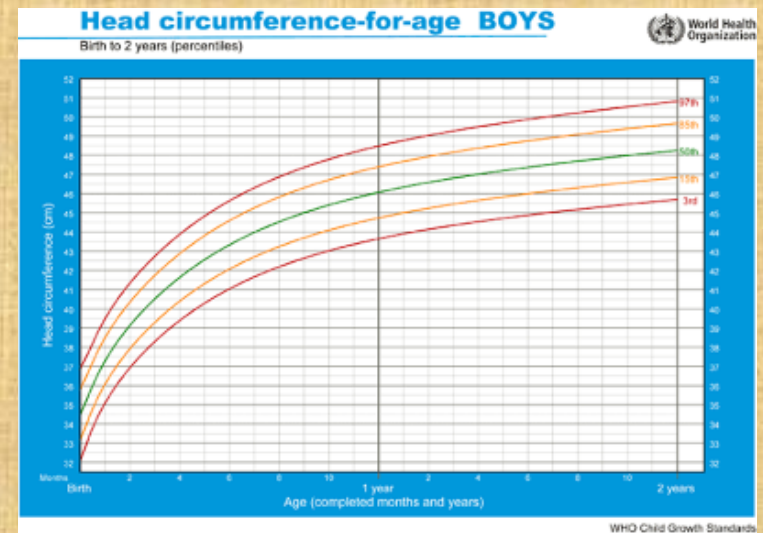
Upper segment: Lower segment ratio

- Birth: 1.7:1
- 3 years: 1.3:1
- 7 years: 1:1
- Adults: 0.9:1

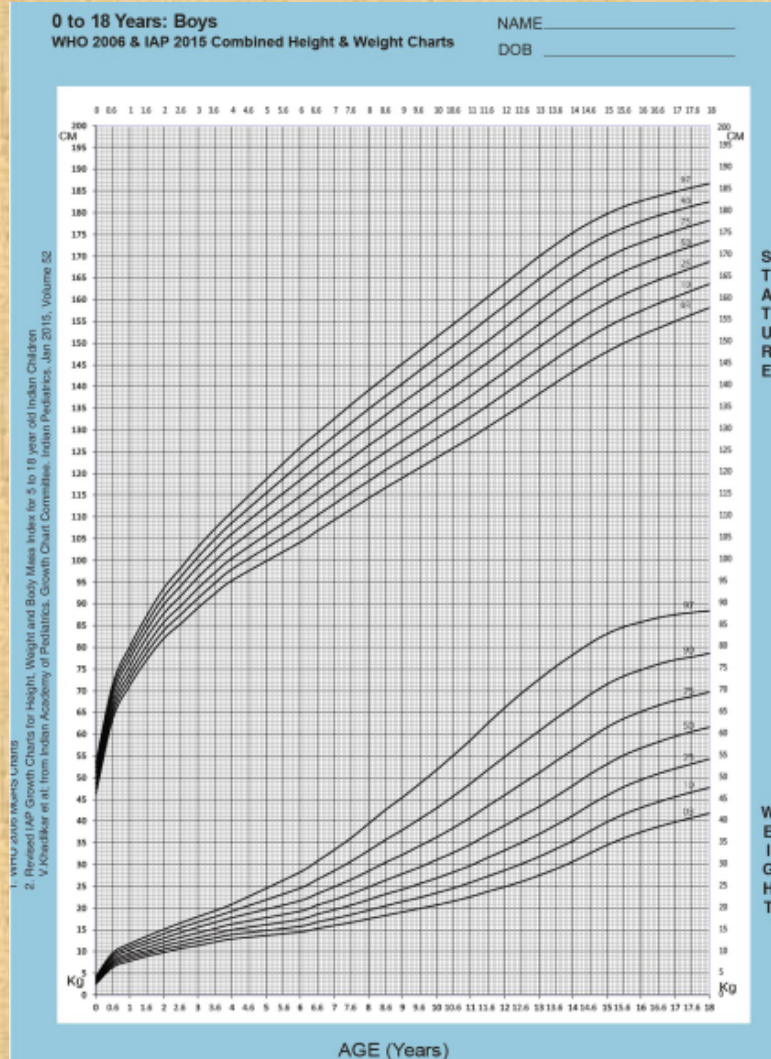
Growth Charts

- IAP charts
- WHO charts
- K N Agrawal charts
- ICMR charts
- CDC charts

WHO Growth Charts



IAP Growth Charts



Development

Principles of Development

1. Continuous process from conception to maturity
2. The sequence of development is the same, but the rate of development varies from child to child
3. Development is intimately related to the maturation of the nervous system
4. Generalized activity replaced by specific responses
5. Development is in the cephalocaudal direction
6. Certain primitive reflexes have to be lost before the voluntary movement is acquired

Domains of Development

- **Gross Motor**
- **Fine Motor**
- **Personal-Social**
- **Language**
- General Understanding (Cognition)
- Vision & Hearing
- Sphincter control

Gross Motor Milestones

- Head holding – 3 months
- Roll over – 5 months
- Sitting with support – 6 months
- Sitting without support – 8 months
- Standing with support – 9 months
- Standing without support – 12 months

Gross motor milestones (contd...)

- Walking without support – 15 months
- Runs – 18 months
- Walks upstairs and downstairs – 2 years
- Rides tricycle – 3 years
- Hops on one foot – 4 years

Fine Motor Milestones

- Reach for objects/Bidextrous approach–4 mo
- Unidextrous approach/Palmar grasp– 5 mo
- Transfer of objects – 6 mo
- Ulnar grasp – 7 mo
- Immature pincer grasp / Radial grasp– 9 mo
- Mature pincer grasp – 12 mo

Fine Motor Milestones (contd...)

- Imitates scribbling – 15 months
- Tower of 3 cubes – 18 months
- Tower of 6 cubes – 2 years
- Tower of 9 cubes / Copies Circle – 3 years
- Copies cross – 4 years
- Copies triangle – 5 years

Language Milestones

- Coos – 2 months
- Babbles/Monosyllables – 6 months
- Bisyllables – 9 months
- 1 to 2 words with meaning – 12 months
- 10 words – 18 months
- 2 word sentences – 2 years
- Tells stories – 3 to 4 years

Social Milestones

- Social smile – 2 mo
- Recognizes mother – 3 mo
- Smiles at mirror image – 6 mo
- Stranger anxiety – 7 mo
- Waves bye-bye – 9 mo
- Comes when called/Simple ball game – 12 mo

Social Milestones (contd...)

- Indicates needs by pointing – 15 mo
- Dry by day – 18 months
- Dry by night – 2 years
- Dresses and undresses – 3 years
- Group play / Does to toilet alone – 4 years
- Independent dressing and bathing – 5 years

Developmental Assessment

- Screening tests:
 - Denver Developmental Screening Test (DDST)
 - Trivandrum Development Screening Test (TDST)
 - Phatak's Baroda Screening Test
- Definitive tests:
 - Bayley Scale of Infant Development (BSID)
 - Developmental Assessment Scale for Indian Infants (DASII)

Equipment for Developmental Assessment

- Red ring tied to string
- Nine red cubes
- Paper pellets
- Spoon
- Cup with handle
- Book with thick pages
- Picture book
- Red pencil, paper
- Bell

Question 1

At what age does the weight of the child double that at birth?

Ans. 1

4 months of age

Question 2

At what age does the Upper segment : Lower segment ratio become 1:1?

Ans. 2

7 years

Question 3

What is a pediatric patient between the age of 1 and 3 years called?

Ans. 3

Toddler

Question 4

At what age does an infant learn to wave
“Bye-Bye?”

Ans. 4

9 months

Question 5

At what age does a child learn to ride a tricycle?

Ans. 5

3 years

Thank you